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**Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025**  
**Data Structures and Applications**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
 2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                               |  | M | L  | C   |
|------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|----|-----|
| Q.1        | a. | Define Data Structures. Explain the classification of data structures with a neat diagram.                                                                                                                    |  | 8 | L2 | CO1 |
|            | b. | Write a C Functions to implement pop , push and display operations for stacks using arrays.                                                                                                                   |  | 7 | L2 | CO2 |
|            | c. | Differentiate structures and unions.                                                                                                                                                                          |  | 5 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                               |  |   |    |     |
| Q.2        | a. | Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression.<br>6 2 / 3 – 4 2 * +.                                                                                |  | 7 | L3 | CO2 |
|            | b. | Explain the dynamic memory allocation function in detail.                                                                                                                                                     |  | 8 | L2 | CO1 |
|            | c. | What is Sparse matrix? Give the triplet form of a given matrix and find its transpose<br>$A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$ |  | 5 | L3 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                               |  |   |    |     |
| Q.3        | a. | Define Queue. Discuss how to represent a queue using dynamic arrays.                                                                                                                                          |  | 8 | L2 | CO2 |
|            | b. | Write a C Function to implement insertion ( ) , deletion ( ) and display ( ) operations on circular queue.                                                                                                    |  | 6 | L3 | CO2 |
|            | c. | Write a note on Multiple stacks and queues with suitable diagram.                                                                                                                                             |  | 6 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                               |  |   |    |     |
| Q.4        | a. | What is a linked list? Explain the different types of linked list with neat diagram.                                                                                                                          |  | 6 | L2 | CO3 |
|            | b. | Write a C function for the following on singly linked list with example :<br>i) Insert a node at the beginning<br>ii) Delete a node at the front<br>iii) Display.                                             |  | 8 | L3 | CO3 |
|            | c. | Write the C function to add two polynomials.                                                                                                                                                                  |  | 6 | L2 | CO3 |

| Module – 3 |    |                                                                                                                                                                 |    |    |     |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.5        | a. | Discuss how binary trees are represented using : i) Assay ii) Linked list.                                                                                      | 6  | L2 | CO4 |
|            | b. | Define Threaded binary tree. Discuss In – threaded binary tree.                                                                                                 | 6  | L2 | CO4 |
|            | c. | Write the C function for the following additional list operation :<br>i) Inverting Singly linked list ii) Concatenating Singly linked list.                     | 8  | L3 | CO3 |
| OR         |    |                                                                                                                                                                 |    |    |     |
| Q.6        | a. | Discuss Inorder , Preorder , Postorder and Level order traversal with suitable function for each.                                                               | 8  | L3 | CO4 |
|            | b. | Define the threaded binary tree. Construct threaded binary tree for the following element : A, B, C, D, E, F, G, H, I.                                          | 6  | L2 | CO4 |
|            | c. | Write a C function for the following :<br>i) Insert a node at the beginning of doubly linked list.<br>ii) Deleting a node at the end of the doubly linked list. | 6  | L3 | CO3 |
| Module – 4 |    |                                                                                                                                                                 |    |    |     |
| Q.7        | a. | Define Forest , Transform the forest into a binary tree and traverse using inorder , preorder and postorder traversal with an example.                          | 8  | L1 | CO5 |
|            | b. | Define Binary search tree. Construct a binary search tree for the following elements : 100 , 85 , 45 , 55 , 120 , 20 , 70 , 90 , 115 , 65 , 130 , 145.          | 6  | L2 | CO5 |
|            | c. | Discuss Selection tree with an example.                                                                                                                         | 6  | L2 | CO5 |
| OR         |    |                                                                                                                                                                 |    |    |     |
| Q.8        | a. | Define Graph. Explain adjacency matrix and adjacency list representation with an example.                                                                       | 8  | L2 | CO5 |
|            | b. | Define the following terminology with example :<br>i) Digraph ii) Weighted graph iii) Self loop iv) Connected graph.                                            | 6  | L2 | CO5 |
|            | c. | Briefly explain about Elementary graph operations.                                                                                                              | 6  | L3 | CO5 |
| Module – 5 |    |                                                                                                                                                                 |    |    |     |
| Q.9        | a. | Explain in detail about Static and Dynamic Hashing.                                                                                                             | 6  | L2 | CO5 |
|            | b. | What is Collision? What are the methods to resolve collision?                                                                                                   | 7  | L2 | CO5 |
|            | c. | Explain Priority queue with the help of an examples.                                                                                                            | 7  | L2 | CO5 |
| OR         |    |                                                                                                                                                                 |    |    |     |
| Q.10       | a. | Define Hashing. Explain different hashing functions with suitable examples.                                                                                     | 12 | L2 | CO5 |
|            | b. | Write short note on :<br>i) Leftist trees ii) Optimal binary search tree.                                                                                       | 8  | L3 | CO5 |

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